



Alternative oxidase in bacteria

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ABSTRACT

While alternative oxidase (AOX) was discovered in bacteria in 2003, the expression, function, and evolutionary history of this protein in these important organisms is largely unexplored. To date, expression and functional analysis is limited to studies in the Proteobacteria *Novosphingobium aromaticivorans* and *Vibrio fischeri*, where AOX likely plays roles in maintenance of cellular energy homeostasis and supporting responses to cellular stress. This review describes the history of the study of AOX in bacteria, details current knowledge of the predicted biochemical and structural characteristics, distribution, and function of bacterial AOX, and highlights interesting areas for the future study of AOX in bacteria.

1. Introduction

Bacteria are found in large numbers in the environment, colonizing a wide variety of habitats and contributing to essential ecosystem processes such as biogeochemical cycling [1]. Part of their success in surviving and thriving in these varying habitats lies in their respiratory flexibility [2]. Compared to eukaryotes, bacteria typically can use a much more diverse range of electron donors and acceptors in their respiratory pathways to allow growth under a wide range of oxygen tensions [3]. For example, a typical eukaryotic mitochondrial aerobic respiratory pathway includes electrons entering the pathway through the activity of NADH dehydrogenase (complex I) or succinate dehydrogenase (complex II), being passed to the quinone pool through a ubiquinone-cytochrome C reductase (cytochrome *bc1* complex or complex III) and terminating at a cytochrome oxidase (complex IV) (Fig. 1A). The activity of complexes I, III, and IV are linked to energy conservation through the generation of the protonmotive force, which can be used to drive the synthesis of ATP. Depending on the eukaryote, the pathway could also include alternative NADH dehydrogenase (s) [4] or the quinol oxidase, alternative oxidase [5], whose activities are typically not directly linked to energy conservation.

On the other hand, bacteria can vary widely in aerobic respiratory chain components. Some, like *Escherichia coli*, lack proteins equivalent to complexes III and IV, instead harboring multiple NADH dehydrogenases and quinol oxidases for aerobic respiration [3]. Others, such as *Vibrio fischeri*, encode proteins corresponding to complexes I-IV and multiple NADH dehydrogenases and quinol oxidases (Fig. 1B). As

more bacterial genomes are sequenced, more variations on these pathways are being discovered. This brings to mind the questions of how individual respiratory pathway components function, how are the various components used to construct pathways that will support the growth and survival of bacteria under varying environmental conditions, and ultimately how the function of these respiratory pathways impact bacterial contributions to ecosystem processes.

One interesting example of genome sequencing contributing to a better understanding of the diversity of aerobic respiratory components in bacteria involves alternative oxidase. Alternative oxidase (AOX) is a diiron quinol oxidase that functions as a terminal oxidase to reduce oxygen to water. This activity is not directly linked to the generation of the proton motive force [7]. Instead, the energy liberated by the activity of AOX is released as heat. AOX is ubiquitous in plants and can be found in certain other eukaryotes [5], including animals [8], fungi [9], microsporidia [10], and protists [11,12]. Examples of described roles for AOX include the generation of heat in thermogenic plants [13], as the terminal oxidase used by the bloodstream form of trypanosomes [14], and a role in growth and stress tolerance in plants experiencing abiotic stress [15].

Until genome sequencing became more common, it was thought that bacteria did not encode AOX. The first reports of the discovery of AOX in bacteria were in *Novosphingobium aromaticivorans* in 2003 [16,17]. Since this initial finding, the study of AOX in bacteria has lagged behind eukaryotes [18]. However, understanding more about the expression and function of this interesting protein in bacteria can provide insight into bacterial physiology and how physiological function impacts ecosystem

Abbreviations: AOX, alternative oxidase; ATP, adenosine triphosphate; NADH, nicotinamide adenine dinucleotide; NO, nitric oxide; PTOX, plastid terminal oxidase.

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processes while also contributing to a more comprehensive comparative analysis of the evolution and function of AOX across the domains of life. This review describes the history of the study of AOX in bacteria, the predicted biochemical and structural characteristics of bacterial AOX, current knowledge about the distribution and function of this protein in bacteria, and highlights future directions for investigation.

2. History of the study of alternative oxidase in bacteria

Although cytochrome-independent alternative oxidase activity had been described in *Neurospora* in the 1950s [19], bacterial genes encoding AOX were not discovered until genomic sequences became available in the early 2000s. In 2003–2004, several papers reported the identification and characterization of an AOX-encoding gene in the genome of the alphaproteobacterium *N. aromaticivorans* [16,17,20]. Finnegan et al. [16], McDonald et al. [20], and Atteia et al. [21] used sequence-based comparisons to propose that the evolutionary history of AOX included prokaryotic origins and radiation through eukaryotes occurred through endosymbiosis events that gave rise to mitochondria. Stenmark and Nordlund [17] heterologously expressed the *N. aromaticivorans* AOX in *E. coli* to demonstrate terminal oxidase activity and used Western blotting to identify culture conditions that impacted the expression of this protein in *N. aromaticivorans*.

The discovery of a functional AOX in *N. aromaticivorans* led to questions about the distribution of this interesting protein among the bacteria. Around this time, along with the increase in genomic sequence data for individual bacterial isolates, large environmental genome sequence (metagenomic) datasets became available. McDonald and Vanlerberghe [22] analyzed a microbial metagenomics dataset from the Sargasso Sea [23]. They discovered that AOX was distributed among Eubacteria found in surface water samples of this nutrient-limited, open ocean environment. Within the ~1800 genomic species estimated to be present in the dataset, 67 putative bacterial AOX proteins were identified using similarity, synteny, and phylogenetic analyses. This apparent abundance of bacterial *aox*-like genes in marine bacteria led to the questions of the role of AOX respiration in bacterial physiology and how bacterial AOX function could ultimately impact ocean ecosystem function and productivity.

A genetically manipulable marine bacterium as a model organism is helpful in answering these questions about the role(s) of AOX in

bacterial physiology and marine ecosystems. Analysis of the genome sequence of *V. fischeri* ES114 [24] revealed the presence of an *aox*-like gene. *V. fischeri* ES114 is a model for studying bioluminescence, quorum (or pheromone) sensing, and mutualistic host-microbe interactions [25]. As a result, numerous genetic tools have been developed for this organism. Using the *N. aromaticivorans* experiments as a guide [17], initial attempts to use a *V. fischeri* *aox* mutant to identify phenotypic differences from wild type in various medium types under differing growth conditions and during symbiotic growth were not successful. A breakthrough came from the results of a whole genome expression analysis of *V. fischeri* ES114 in the presence and absence of nitric oxide (NO) [26], a molecule of known importance in the symbiotic lifestyle of *V. fischeri*. Although this study focused on the symbiotic role of the NO-sensing protein H-NOX, the dataset revealed that *aox* transcript levels increased in response to NO, suggesting a role for AOX in *V. fischeri*'s response to host-derived NO. Further work, described in more detail below, identified that the regulatory protein NsrR is involved in *V. fischeri* AOX expression [27] and that under laboratory growth conditions, *V. fischeri* AOX can reduce signs of cellular stress when cells are exposed to NO. [28]

Since these initial discoveries of bacterial AOX distribution and function, the amount of sequence data has increased exponentially. In addition, studies of AOX structure and function in other groups of organisms has provided interesting insight into how this protein functions in eukaryotes. While the study of bacterial AOX has lagged considerably, these new data provide an opportunity to approach bacterial AOX with fresh eyes and tools to advance understanding of how this interesting protein has evolved and functions in this important group of organisms.

3. Predicted biochemical and structural characteristics of bacterial AOX

From the wealth of experimentation on eukaryotic AOX proteins, including sequence homology and structural comparisons, general conclusions have been made about the universal characteristics of AOX proteins. As a whole, AOXs are diiron proteins that function as quinol oxidases to reduce oxygen to water without directly translocating protons across a membrane [7]. The crystal structure of trypanosomal AOX has been determined [29], and sequence analysis along with homology modeling has revealed commonalities between AOX proteins from

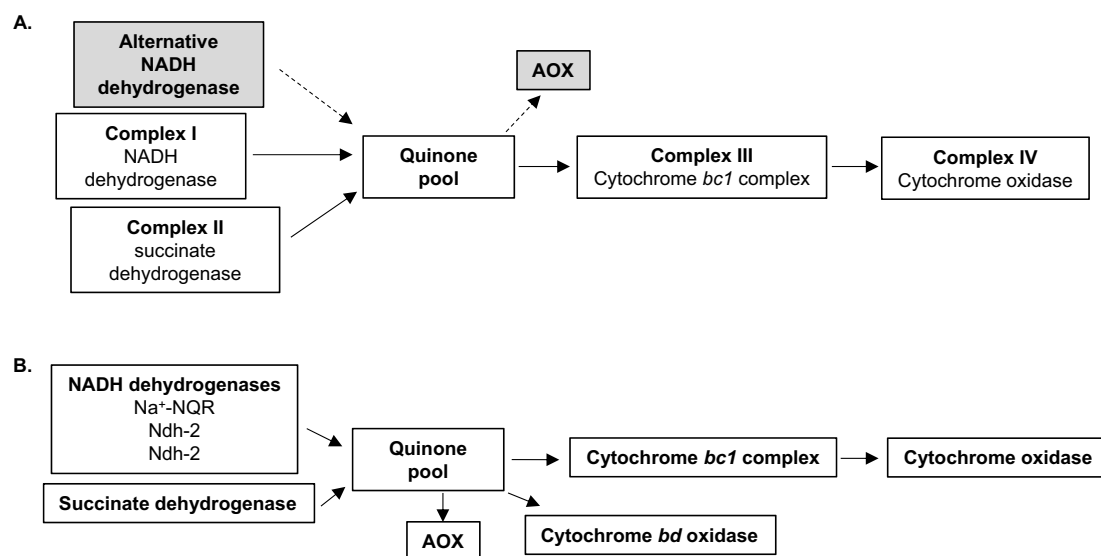


Fig. 1. Examples of aerobic respiratory pathways. A. A typical eukaryotic mitochondrial aerobic respiratory pathway. Shaded boxes represent components that are found only in certain eukaryotes. A few additional enzymes that can catalyze reduction of the quinone pool are not shown [6]. B. Aerobic respiratory pathway components present in the bacterium *Vibrio fischeri*. Na⁺-Nqr = sodium translocating NADH dehydrogenase, Ndh-2 = alternative NADH dehydrogenase, and the *V. fischeri* genome encodes two Ndh-2 enzymes.

representatives from different kingdoms of life including predictions that all AOXs exist as homodimers and associate with inner membranes (reviewed in [30]). In addition, these analyses have identified fully conserved amino acid residues associated with the diiron center, highly conserved residues associated with quinol interactions and structure, and conserved hydrophobic residues proposed to be important in membrane association [31]. While these comparative analyses are in their infancy with bacterial AOX, several interesting similarities and differences to other AOXs have been noted.

Sequence analyses have revealed that bacterial AOXs contain many of the conserved amino acid residues known to be important for AOX

structure and function. However, there are differences in protein length and overall amino acid composition [5,16,30,31]. Despite the noted amino acid residue conservation, differences in the activity of eukaryotic and bacterial AOXs have been described. For example, the trypanosomal AOX and *N. aromaticivorans* AOX share many similarities in important amino acid residues. Yet, the purified proteins vary dramatically in their measured ubiquinol oxidation activities, with *N. aromaticivorans* AOX displaying a much lower $V_{\max}$ value [30]. Homology modeling of the hydrophobic cavity and diiron core of the *N. aromaticivorans* AOX indicates that there are likely differences in the architecture of the cavity leading to the active site that could help explain the noted differences in

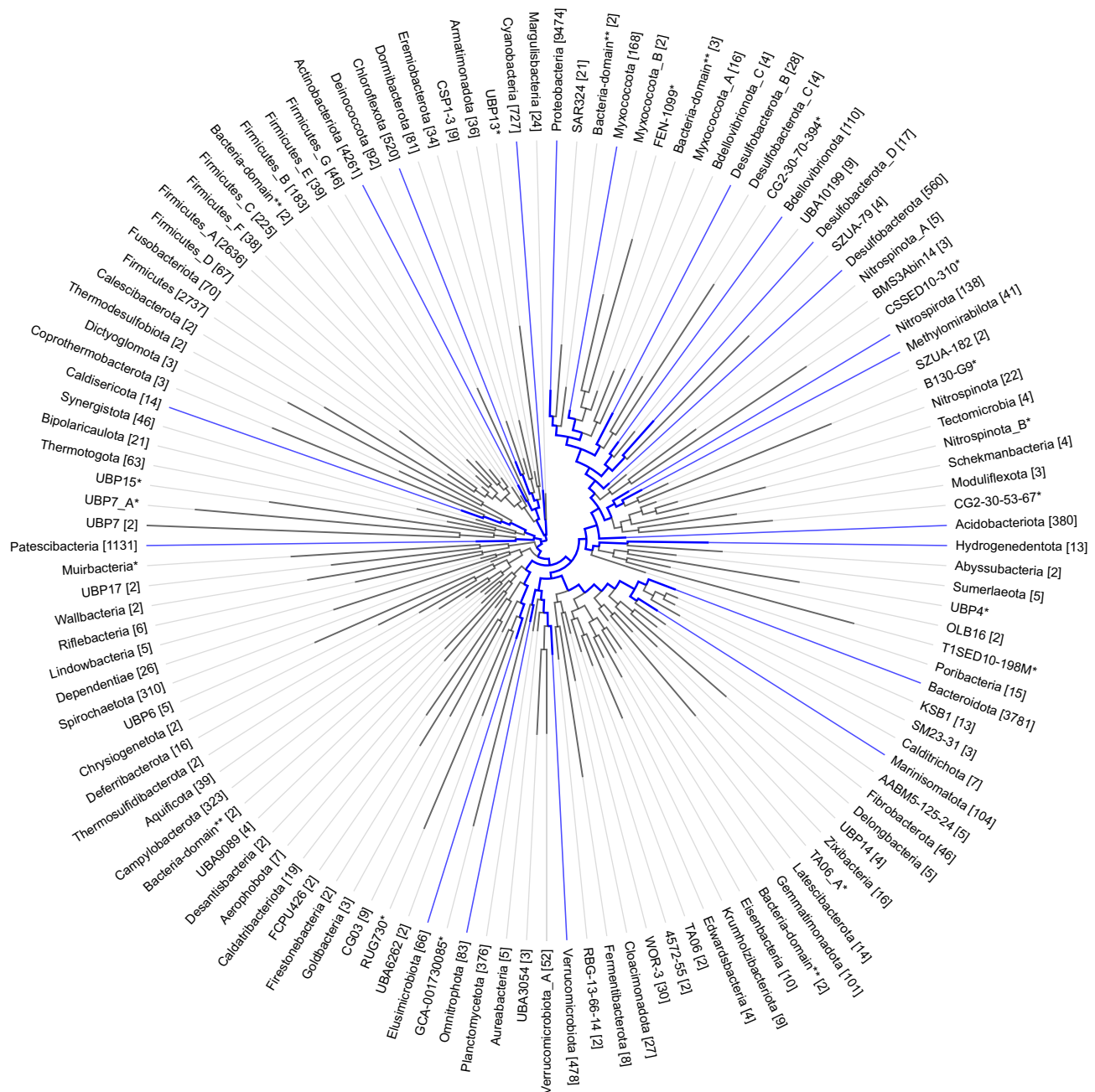


Fig. 2. Phylogenomic distribution of the Pfam AOX protein family (PF01786) in bacterial phyla indicates a broad but uneven distribution of AOX among the bacteria. The tree was generated using AnnoTree [33], searched with PF01786 with an E value of 0.00001 in July 2022. Blue highlights indicate phyla where at least one genome sequence contains a hit to PF01786. The number in brackets shows the number of genomes available for query in the clade. * = only one genome available for query ** = no corresponding taxonomy at the phylum level; uses a higher-level taxonomy instead. The PF01786 Pfam AOX protein family also includes the cyanobacterial plastid terminal oxidase (PTOX); therefore, the positive result in the Cyanobacteria phylum does not necessarily indicate the presence of AOX. Table 1 provides information on the number of genomes within each phylum with PF01786 hits.

activity [30]. Recently, a sequence, structural, and functional comparison of Arabidopsis and trypanosomal AOX was reported [32]. The results of this study suggest that differences in sequence and structure can be connected to differences in the kinetics and sensitivity of each enzyme to inhibitors. Expanding these types of studies to bacterial systems can provide interesting insight into the overall function of bacterial AOX, how the function and activity of bacterial AOXs compare to eukaryotic AOXs, and whether there is predicted variability in function and activity within bacterial AOXs.

4. Distribution of alternative oxidase-like genes in bacteria

The initial studies of bacterial AOX in early genomic and metagenomic datasets suggested that this protein was mainly encoded by Proteobacteria living in marine environments [5,22], although also occasionally found in Proteobacteria like *N. aromaticivorans* [17] isolated from terrestrial environments. However, since then, the amount of sequence data has increased considerably and revealed a broader yet uneven distribution of AOX in bacteria. To illustrate, AnnoTree [33], which combines functional annotation data with evolutionary data for over 27,000 bacterial genome sequences, was searched for the phylogenomic distribution of the Pfam [34] AOX family PF01786 (Fig. 2, Table 1). The search revealed 987 genomes with annotations associated with PF01786, but these include 220 hits found in the Cyanobacteria (Table 1) that most likely correspond to the cyanobacterial plastid terminal oxidase (PTOX [35]); PTOX is included in PF01786. Most bacteria encode a single copy of *aox*, although a few exceptions have been noted [18]. AnnoTree failed to identify annotations associated with PF01786 in any of the ~1500 available archaeal genomes.

While most of the genomes with AOX annotations identified by AnnoTree do fall within the Proteobacteria, there is a diversity of phyla where AOX can be found (Table 1). Within these phyla, many of the bacteria with AOX are associated with marine environments, although there are examples of bacteria isolated from freshwater or terrestrial environments. These include both nonpathogenic and potentially pathogenic strains (*Legionella* spp., *Vibrio* spp., etc.). Notably, some of the groups of bacteria whose genomes appear to encode AOX include members of numerically dominant and ecologically relevant marine proteobacterial groups such as SAR11 (~25 % of all plankton [36]), SAR86 [37], SAR116 [38], and the *Roseobacter* clade (~20 % of coastal bacterial communities [39]).

The distribution and abundance of AOX annotations within bacterial

Table 1
Listing of AnnoTree phylum-level distribution results of AOX representing blue highlighted lineages from Fig. 2.

Phylum name	Total number of genomes searched	Total number of genomes with PF01786 hits
Proteobacteria	9474	630
Myxococcota	168	1
Desulfobacterota_B	28	1
Bdellovibrionota	110	1
Desulfobacterota_D	17	1
Desulfobacterota	560	3
Nitrospirata	138	2
Methylomirabilota	41	11
Acidobacteriota	380	1
Hydrogenedentota	13	2
Bacteroidota	3781	25
Marinisomatota	104	4
Verrucomicrobiota	478	1
Omnitrophota	83	4
Elusimicrobiota	66	3
Patescibacteria	1131	21
Caldisericota	14	1
Actinobacteriota	4261	50
Chloroflexota	520	5
Cyanobacteria	727	220

genomes are interesting in several ways. First, while there is a broad diversity of phyla, the total number of phyla with genomes with AOX annotations and the total number of AOX annotations is not remarkably high. As shown in Table 1, AnnoTree analysis identified 987 genomes distributed among 20 phyla. For comparison, using AnnoTree to perform a similar search using PF01654, which corresponds to subunit I of the cytochrome *bd* quinol oxidase [40], results in 16,193 positive genomes distributed among 74 phyla (data not shown). In addition, looking more closely at the taxonomic distribution of AOX annotations, they are not universally found within members of a particular genus. As an example, in the 157 *Vibrio* spp. genomes analyzed by AnnoTree, 35 were positive for an AOX annotation. While AnnoTree does not query all available genomic and metagenomic datasets, the results suggest exploring the evolutionary history of AOX in bacteria is warranted.

5. Functional roles for bacterial alternative oxidase

The first report of a functional AOX in bacteria was in *N. aromaticivorans* DSM 12444 [17]. *N. aromaticivorans* DSM 12444 is a proteobacterium isolated from subsurface sediments and can notably use aromatic hydrocarbons as a sole carbon and energy source as well as sugars such as glucose or lactate [41]. With the availability of a complete genome sequence for analysis, it was recognized that the genome of this strain encoded a putative AOX [16,17,20]. To determine whether it was a functional AOX, Stenmark and Nordlund [17] heterologously expressed this protein in an *E. coli* strain lacking the ability to respire aerobically due to the lack of terminal respiratory oxidases. Expression of *N. aromaticivorans* AOX restored the ability of the *E. coli* strain to grow aerobically, demonstrating function as an oxidase. This experiment was followed with the Western blotting of proteins isolated from *N. aromaticivorans* liquid cultures grown in minimal medium with different carbon sources and agitation levels to identify whether AOX was expressed in *N. aromaticivorans* and, if so, whether culture conditions impacted the levels of the protein present in cells. They found that AOX levels were highest when cultures were less agitated (lower oxygen level) and when glucose rather than lactate was provided as a carbon source [17].

N. aromaticivorans requires oxygen to grow but grows best under microaerobic conditions in minimal medium containing glucose or lactose as a carbon source [41]. Based on this information and the results of the Western-blotting experiments, Stenmark and Nordlund [17] hypothesized that the function of *N. aromaticivorans* AOX is not solely to scavenge oxygen but instead to adapt to imbalances between carbon metabolism and electron transport. Unfortunately, this hypothesis has not yet been experimentally tested, nor has the expression of *N. aromaticivorans* AOX been explored during growth with other carbon sources. However, these results bring to mind interesting questions about how oxygen tension and energy flux in bacterial cells might connect to AOX function.

The only other bacterium in which experiments involving AOX expression and function have been published is the marine bacterium *V. fischeri* ES114 [27,28]. As mentioned above, *V. fischeri* is a model bacterium that is genetically tractable and whose genome encodes AOX along with two other respiratory oxidases (Fig. 1B). Similar to the *N. aromaticivorans* AOX study [17], *V. fischeri* AOX was shown to function as a terminal oxidase and allow an *E. coli* aerobic respiratory mutant to grow in the presence of oxygen. However, unlike *N. aromaticivorans*, *V. fischeri* *aox* was not expressed under aerobic conditions in a standard nutrient-replete laboratory medium. Instead, *V. fischeri* *aox* expression is induced in response to exogenously provided NO through the activity of the regulatory protein NsrR. Using various *V. fischeri* respiratory mutant strains along with respirometry and NADH oxidation assays, it was shown that the activity of *V. fischeri* AOX is more resistant to inhibition by NO than the *V. fischeri* cytochrome *bd* oxidase CydAB, the primary oxidase used by *V. fischeri* cells under normal aerobic laboratory growth conditions [27]. Therefore, *V. fischeri* AOX is expressed in response to NO

and likely functions to benefit the cells when the activity of other respiratory oxidases is compromised. Interestingly, *V. fischeri* AOX can functionally complement the activity of the cytochrome *bd* oxidase, CydAB. *V. fischeri* strains lacking the proton motive force-generating CydAB (Fig. 1B) display a severe aerobic growth defect on rich medium, and suppressor mutants frequently arise that restore aerobic growth. Analysis of randomly selected suppressor mutants indicated that the restoration of aerobic growth was due to disruptions in the activity of NsrR, effectively producing cells that constitutively expressed AOX [27]. While it is still unclear under these experimental conditions which respiratory dehydrogenases are coupled with AOX activity and whether the other cytochrome oxidase is active, these results suggest that the benefit of AOX function that allows growth in the suppressor mutants is related to either reducing cellular stress or restoring metabolic homeostasis. Further work with these mutants will likely provide additional insight into AOX function in *V. fischeri*.

Unanswered questions from the work described above include why *V. fischeri* AOX is not expressed during growth in laboratory medium unless exogenous NO is provided and whether this regulation of AOX expression is related to the conditions *V. fischeri* encounters in the environment. As *V. fischeri* is exposed to NO when colonizing the eukaryotic squid host, *Euprymna scolopes* [42], there is an ecologically-relevant situation where the cells could encounter NO. However, no phenotypic difference between an *aox* mutant and wild-type *V. fischeri* has been found during the establishment of the symbiosis with *E. scolopes*. While it is possible that the benefit of AOX function is primarily during a lifestyle outside of the host, another possibility is that AOX may provide physiological benefits for *V. fischeri* that are not easily observed as measurable phenotypes during host colonization.

While the work described above identified a connection between NO and the expression and activity of *V. fischeri* AOX (Fig. 3), an *aox* mutant did not display any identifiable growth defects compared to wild type under the laboratory growth conditions that resulted in expression of *aox*. To increase the impact of exogenously added NO on *V. fischeri* cells, a strain lacking a major aerobic NO detoxification enzyme,

flavo-hemoglobin or Hmp [43], was constructed, and the *aox* deletion mutation was introduced into the *hmp*-parental strain. In these strains lacking Hmp, it became apparent that the presence of AOX does positively impact the growth of the cells in mineral salts medium with exogenously-applied NO. [28] Transcriptome analysis along with assays to measure reactive oxygen species and membrane integrity indicated that the absence of functional AOX during growth with NO increased signs of cellular stress under the growth conditions tested. One possible explanation of these results is that a role of AOX could be to allow the function of the aerobic electron transport chain when the activity of the other oxidases is impaired, resulting in lower NADH levels, reactive oxygen species, and cellular stress [28].

These results in *V. fischeri* raise interesting questions. There is a demonstrated link in this bacterium between NO, AOX, and indicators of cellular stress [27,28]. However, these experiments relied on controlled laboratory growth conditions with exogenously applied NO and, in some instances, selective use of mutant backgrounds. While these approaches are beneficial for helping to understand cellular processes, they do not necessarily reflect the conditions bacteria encounter in their natural habitats. For example, it is known that *V. fischeri* encounters NO during the host-associated lifestyle [42]. Still, it is unclear how well the growth conditions and NO levels used in the lab studies relate to what the bacteria are exposed to during host colonization. These parameters have not yet been quantitatively measured in the host. In addition, very little is known about *V. fischeri*'s lifestyle outside of the host and whether cells could be exposed to NO through the activity of denitrifying bacteria [44] or nitrite photolysis in seawater [45], and how this might relate to AOX expression and function. There are exciting opportunities to discover more about *V. fischeri* AOX in laboratory-based studies, to better connect these to the environments *V. fischeri* has evolved in and to use new approaches and techniques to investigate AOX expression and function in natural environments.

The studies in *N. aromaticivorans* [17] and *V. fischeri* [28] described above indicate that benefits of AOX function in bacteria include maintenance of cellular energy homeostasis and supporting responses to

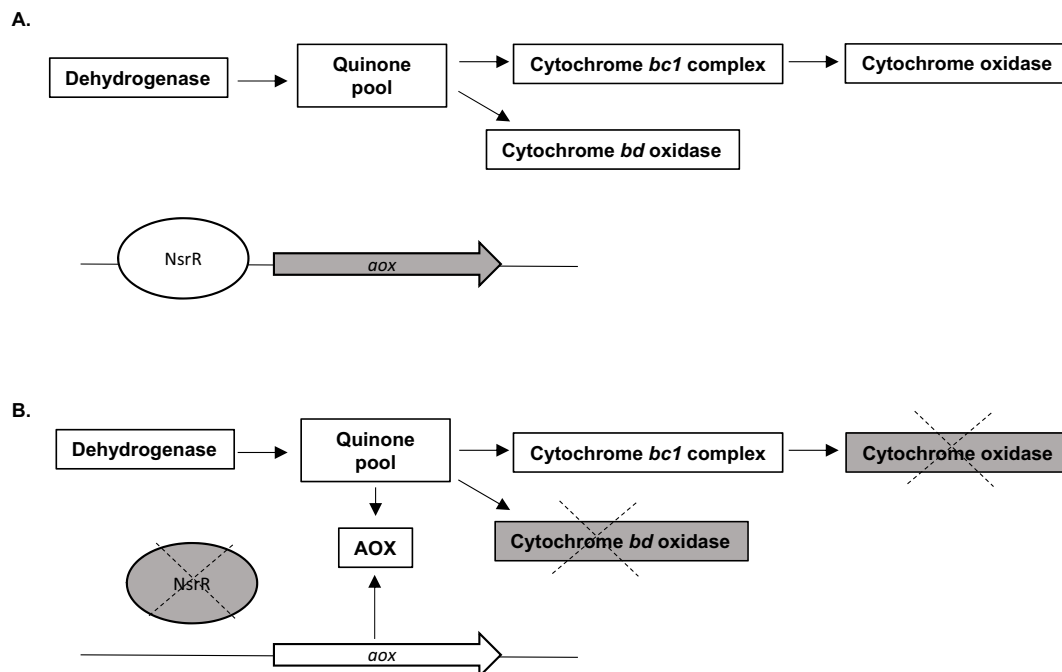


Fig. 3. Summary of the currently known expression patterns and function of AOX in *V. fischeri*. A. Under normal laboratory growth conditions, the cytochrome oxidases are expressed, but *aox* expression is repressed by the NO-responsive negative regulator NsrR. B. When exogenous NO is provided to cells grown under normal laboratory conditions, the activity of the cytochrome oxidases is inhibited and NO binds to NsrR and prevents repression of *aox* expression. AOX is produced and can act as a quinol oxidase. During NO exposure, the activity of AOX is associated with reduced signs of cellular stress. It is currently unclear which specific dehydrogenases are coupled to the activity of the various oxidases.

stress. Although more work is needed to explore the expression patterns and physiological role(s) of bacterial AOX, these results are compatible with observations in plants [15], fungi [46], diatoms [47], and animals [48], and suggest that in bacteria that encode AOX, this protein plays a key role in their physiology, growth, and survival

6. Future directions

Despite the importance of bacteria in the functioning of our planet, our knowledge of how different bacterial respiratory pathways and proteins function to promote the growth and survival of bacteria and ultimately influence ecosystem processes is far from complete. One excellent example is bacterial AOX. Learning more about the expression and function of AOX in bacteria, especially those that are abundant in the environment and known to play key roles in processes such as biogeochemical cycling, will not only provide general insight into bacterial physiology and the evolutionary history of AOX, but also the impact that bacterial physiological pathways can have on broader ecosystem-level processes. As mentioned throughout this review, there is much to be learned about bacterial AOX, and many opportunities for interesting avenues of research exploration on multiple scales. A few of these are highlighted below.

In general, there is a knowledge gap about the expression patterns of AOX in diverse bacteria and how AOX activity is coupled to that of other potential respiratory chain components. As mentioned above, *N. aromaticivorans* AOX expression levels changed in response to the carbon source provided to the cells [17], whereas in *V. fischeri*, *aox* expression was not impacted by the carbon source of the medium but instead induced in response to NO. [27] It has not been determined in either of these organisms which enzymes in the electron transport chain are coupled to AOX activity. In *V. fischeri*, data suggests that an NADH dehydrogenase is acting to reduce the quinone pool when AOX is functional, but this has not been directly measured [28]. Current efforts to address whether there are common patterns in AOX expression across bacteria and whether they couple the activity of specific respiratory proteins with AOX involve studying this protein in other genetically tractable bacteria that represent a greater diversity of taxonomic lineages. This approach is being complemented by exploration of metagenomic and metatranscriptomic datasets to learn more about the distribution and expression patterns of *aox* in various environments, especially in bacteria that have yet to be cultured or lack genetic tools. Together, these types of studies will provide a broader picture of the expression patterns of AOX in diverse bacteria and identify possible links between environmental conditions and *aox* expression that can drive hypothesis generation for future studies.

In terms of bacterial AOX function, there is obviously limited data, but the work in *N. aromaticivorans* and *V. fischeri* indicates that physiological benefits of AOX function in bacteria include maintenance of cellular metabolic homeostasis [17] and responses to stress [28]. More direct tests of these roles for AOX, both in these bacteria and in others, is warranted, especially under growth conditions that more closely mimic those found in the environments these bacteria inhabit (nutrient concentrations, levels of NO, etc.). One notable area for further exploration is the connection between reduced oxygen levels and increased AOX content that was described for *N. aromaticivorans* [17]. This appears counterintuitive, as AOX enzymes are typically expected to have a lower affinity for oxygen than other respiratory oxidases and therefore be expected to be used by cells when oxygen levels are higher.

While the affinity of bacterial AOXs for oxygen has not been directly measured, it is possible that they could be similar to that of plants (K_m of $\sim 2\text{--}20\ \mu\text{M}$ [49–51]) or trypanosomes (K_m of $\sim 1\ \mu\text{M}$ [32]). *N. aromaticivorans* DSM 1244 is like *V. fischeri* ES114 (Fig. 1B) in that the genome also encodes a cytochrome *bd* ubiquinol oxidase (CydB) and a *cbh3* cytochrome oxidase (CcoNOQP) (data not shown). It is predicted that these oxidases would have oxygen affinities similar to those measured in other bacteria (*E. coli* CydB A K_m of $\sim 3\text{--}8\ \text{nM}$ [52] and

Bradyrhizobium japonicum CcoNOQP A K_m of $\sim 5\text{--}8\ \text{nM}$ [53]), and an order of magnitude higher than AOX. While the oxygen tension in the cultures in the *N. aromaticivorans* experiments was not directly measured [17], based on the conditions reported, it would be expected that the concentration would be much lower than atmospheric levels (micro-aerobic or less than $10\ \mu\text{M}$ oxygen [54]). This work with *N. aromaticivorans* did not explore the expression patterns of CydB or CcoNOQP. However, a general assumption is that these higher affinity oxidases would be preferentially used during growth under reduced oxygen as compared to a lower affinity oxidase such as AOX [55]. Interestingly, there are several recent reports that suggest that this general assumption is not always valid and that AOX may function at reduced oxygen tensions despite having a lower measured affinity for oxygen. There is evidence in bacteria that certain other lower affinity bacterial oxidases are expressed and function under reduced oxygen tensions [56]. In addition, the study of AOX in plants has revealed that AOX is functional at oxygen tensions nearing hypoxia [57]. Therefore, studying bacterial AOX expression and function at reduced oxygen tensions that the bacteria may encounter in natural environments may provide useful insight into the role(s) of AOX in bacterial physiology.

Finally, an underexplored area of bacterial AOX research involves evolutionary history. As described above, the distribution of AOX in bacterial lineages is broad (found in 20 phyla), but uneven (found in 987 of the 22,904 genomes in these phyla) (Fig. 2, Table 1), and AOX is not yet found in archaea. Previous analysis of sequences present in databases suggested that AOX and PTOX diverged from a common prokaryotic ancestral protein, and then through endosymbiosis events related to the acquisition of mitochondria and chloroplasts, was transferred to eukaryotes [5,16]. Alternatively, it was proposed that AOX was acquired by bacteria through horizontal gene transfer from plants [17]. A later analysis identified AOX only in Proteobacteria, and it was suggested that AOX was present in a common bacterial ancestor, but then was lost through deletion during evolution in certain lineages [18]. Since these ideas were put forth, the amount of sequence data has increased substantially, and it is now known that AOX can be found in phyla other than the Proteobacteria (Fig. 2, Table 1), although there are no immediately obvious patterns to explain why some taxonomic groups have members with *aox*-like genes and others do not. There are more robust data sets and analysis tools available now than when the original hypotheses were proposed, and it will be interesting to apply these along with information from functional analyses to learn more about the evolutionary history of AOX in bacteria and how this relates to the presence of AOX in other domains of life.

While the current information about bacterial AOX is sparse, it is becoming apparent that this interesting protein can be found in a wider diversity of bacteria than originally thought and that these bacteria can often play important roles in ecosystem processes. As the function of respiratory pathways underlies the growth and survival of bacteria, understanding the expression and function of respiratory proteins like AOX has important implications. Beyond a basic interest in learning more about bacterial physiology and AOX in general, further exploration of the expression, function, and evolutionary history of bacterial AOX has the potential to help form a better understanding of how bacterial physiological pathways influence larger-scale processes in the environment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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